LESSON PLAN

HELICOPTER THEORY (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	17
Total No. of Lesson Plans	27	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Flight control surface balancing
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Flight control surface balancing. Flow Separation Stalling, behaviour of aircraft before and after stall. Factors effecting stalling speed, high incidences stall, high speed stall, G stall, wing tip stall, Basic stalling speed.
5. Specific Objectives	11.1.2.1 Trainee will be able to describe various types of aerofoil sections, conditions of performance, theories, and concept related with performance

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of IB diagrams regarding control surface balancing; Instructor supervision during Q&A; session
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of flight control surfaces?
(b) Define the term 'stalling speed' in your own words.
(c) List three factors that influence stalling speed.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Flow separation and stalling mechanism • Aircraft behavior before and after stall • Basic stalling speed concepts	L & Ds	PPT, IB	15 min
(b)	• Factors affecting stalling speed • High incidence and high speed stalls • G stall and wing tip stall characteristics	L & Ds	PPT, PDF	20 min
(c)	• Control surface balancing theory • Impact of imbalance on flight performance • Preventive measures for control surface issues	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a high-speed stall differs from a low-speed stall.
(b) Describe the role of control surface balancing in maintaining flight stability.
(c) Identify two conditions that lead to a wing tip stall.

12. Home Assignment and Follow Up	Prepare a brief note on the power available and required for jet and propeller aircrafts, linking it to stalling characteristics discussed in this lesson, to be covered in LP 18.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Level flight performances, power available and required for jet and propeller aircrafts. Best climb speeds.	Refer to the master syllabus reference list	LP 18

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Instructor, P No. 12345	Senior Instructor, Chief Instructor, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min